

4. The following tables are maintained by a book dealer.

AUTHOR (author-id:int, name: string, city: string, country: string)

PUBLISHER (publisher-id: int, name: string, city: string, country: string)

CATALOG (book-id: int, title: string, author-id#: int, publisher-id#: int,
category-id#: int, year: int, price: int)

CATEGORY (category-id: int, description: string)

ORDER-DETAILS (order-no: int, #book-id: int, quantity: int)

- i. Create the above tables by properly specifying the primary keys and the foreign keys.
- ii. Enter at least five tuples for each relation.
- iii. Give the details of the authors who have 2 or more books in the catalog and the price of the books is greater than the average price of the books in the catalog.
- iv. Find the author of the book, which has maximum sales.
- v. Demonstrate how you increase the price of books published by a specific publisher by 10%.

```
CREATE TABLE AUTHOR (  
  author_id INT PRIMARY KEY,  
  name VARCHAR(50),  
  city VARCHAR(50),  
  country VARCHAR(50)  
);
```

```
CREATE TABLE PUBLISHER (  
  publisher_id INT PRIMARY KEY,  
  name VARCHAR(50),  
  city VARCHAR(50),  
  country VARCHAR(50)  
);
```

```
CREATE TABLE CATEGORY (  
  category_id INT PRIMARY KEY,  
  description VARCHAR(50)  
);
```

```
CREATE TABLE CATALOG (  
  book_id INT PRIMARY KEY,  
  title VARCHAR(100),  
  author_id INT,  
  publisher_id INT,  
  category_id INT,  
  year INT,  
  price INT,  
  FOREIGN KEY (author_id) REFERENCES AUTHOR(author_id),  
  FOREIGN KEY (publisher_id) REFERENCES PUBLISHER(publisher_id),
```

```
FOREIGN KEY (category_id) REFERENCES CATEGORY(category_id)
);
```

```
CREATE TABLE ORDER_DETAILS (
    order_no INT,
    book_id INT,
    quantity INT,
    PRIMARY KEY (order_no, book_id),
    FOREIGN KEY (book_id) REFERENCES CATALOG(book_id)
);
```

INSERT INTO AUTHOR VALUES

```
(1,'Ravi','Delhi','India'),
(2,'John','London','UK'),
(3,'Meena','Mumbai','India'),
(4,'Smith','New York','USA'),
(5,'Anil','Pune','India');
```

INSERT INTO PUBLISHER VALUES

```
(1,'Pearson','Delhi','India'),
(2,'McGraw','NY','USA'),
(3,'Oxford','London','UK'),
(4,'Cambridge','London','UK'),
(5,'TechBooks','Mumbai','India');
```

INSERT INTO CATEGORY VALUES

```
(1,'Programming'),
(2,'Database'),
(3,'Science'),
(4,'Math'),
(5,'Fiction');
```

INSERT INTO CATALOG VALUES

```
(101,'Python',1,1,1,2020,500),
(102,'SQL Basics',2,2,2,2019,400),
(103,'Physics',3,3,3,2018,600),
(104,'Algebra',4,4,4,2021,550),
(105,'Story Book',5,5,5,2022,300),
(106,'Advanced Python',1,1,1,2021,700),
(107,'Python Projects',1,1,1,2022,650);
```

INSERT INTO ORDER_DETAILS VALUES

```
(1,101,5),
(2,102,3),
```

```
(3,103,7),  
(4,104,2),  
(5,101,4);
```

-- 3. Authors with >=2 books & price > average

```
SELECT A.*  
FROM AUTHOR A  
JOIN CATALOG C ON A.author_id = C.author_id  
WHERE C.price > (SELECT AVG(price) FROM CATALOG)  
GROUP BY A.author_id, A.name, A.city, A.country  
HAVING COUNT(C.book_id) >= 2;
```

-- 4. Author of book with maximum sales

```
SELECT A.*  
FROM AUTHOR A  
JOIN CATALOG C ON A.author_id = C.author_id  
WHERE C.book_id = (  
    SELECT book_id FROM ORDER_DETAILS GROUP BY book_id  
    ORDER BY SUM(quantity) DESC LIMIT 1  
);
```

-- 5. Increase price by 10% for a specific publisher

```
UPDATE CATALOG  
SET price = price + (price * 0.10)  
WHERE publisher_id = 1;
```

